

ANUSHA VENKATESH

Chicago, IL | +1 (312) 843-2409 | venkateshanusha2002@gmail.com | <https://anusha-venkatesh.netlify.app/>
<https://www.linkedin.com/in/anusha--venkatesh/> <https://github.com/anusha-2002>

EDUCATION

Illinois Institute of Technology | Chicago, IL, United States Aug'24 – May'26

Master of Science, Computer Science, **Specialization:** Data Science & Analytics **GPA:** 3.7/4.0

Coursework: *Advanced Database, Data Preparation & Analysis, Big Data Technologies, Machine Learning, Information Retrieval, Software Project Management*

Bangalore Institute of Technology | Bangalore, KA, India Dec'20 – Jun'24

B.E., Information Science and Engineering **GPA:** 3.49/4

Coursework: *Object-oriented programming, Data Structures, Algorithms, Database Management, Software Testing*

SKILLS

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|----------------------------|---|
| • Certification | AWS Certified Cloud Practitioner (CLF-C02)- Amazon Web Services |
| • Programming | Java, Python, R, SQL, JavaScript, HTML5, CSS, React |
| • Data Science & Analytics | Data wrangling, statistical analysis, exploratory data analysis(EDA), JSON, XML |
| • Machine Learning | Pandas, NumPy, Scikit-learn, Jupyter Notebook, ARIMA, LSTM, GARCH |
| • GenAI & NLP | OpenAI API (GPT-4), LangChain, spaCy, Claude, Cursor, RAG, LLMs |
| • Cloud & Data Engineering | AWS (EC2, S3, RDS, Redshift, IAM, VPC, Lambda, Glue), ETL pipelines |
| • Tools & DevOps | Git, Jira, Excel, Power BI, Tableau, CI/CD, REST APIs, Bash, Docker, Postman |
| • Soft Skills | Communication, Problem Solving, Strategic, Innovation, Ownership, Collaboration |

EXPERIENCE

Software Engineering Intern, Alervio Inc, NJ, United States **Sept'25 – Present**

- Developed an NLP-driven **Scan Menu** pipeline with **OpenAI**, **LangChain**, and **spaCy** to extract and classify allergen entities from semi-structured (JSON) menu data, improving detection accuracy from **61% to 87%**.
- Building a **TypeScript**, **React**- based recommendation system suggesting restaurants based on users' allergens and dietary preferences, enabling stable user-facing functionality by cutting onboarding time by **42%**.

Machine Learning Intern, Princeton Smart Engineers, Bangalore, India **Jun – Aug'23**

- Implemented **decision tree** and **random forest classifiers** for a campus recruitment prediction system, performing data processing to address data quality issues and to ensure reliable model training.
- Built and optimized the model performance with **GridSearchCV** hyperparameter tuning, boosting accuracy to **93.7%** and improving candidate selection insights.

PROJECTS

NVIDIA Stock Analysis – Time Series & Risk Modeling **May– Aug'25**

- Designed a data pipeline to analyze NVIDIA stock using **ARIMA**, **LSTM**, and **GARCH**, performing EDA, trend forecasting, volatility modeling, and seasonality detection.
- Improved predictive accuracy with a hybrid **ARIMA-GARCH** model, reducing **RMSE** and identifying clustered volatility patterns for actionable risk and investment insights.

ModuMart: Scalable Microservices E-Commerce Backend **Aug– Dec'24**

- Built a scalable microservices e-commerce back-end using **Spring Boot**, **Kafka** for real-time communication, and **Eureka** for service discovery to handle product and order management efficiently.
- Containerized **microservices** with **Docker** and implemented a centralized **API Gateway** for secure, efficient routing, resulting in improved system uptime and a 30% reduction in inter-service latency under load.

Socio-economic Status Classification for Poverty Mitigation **Jan- May'24**

- Developed an image-based poverty classification model using **satellite imagery** and **census data**; applied **ResNet50 CNN** with **Gaussian blur** preprocessing to classify cities into high, medium, and low poverty levels.
- Optimized model performance through hyperparameter tuning and cross-validation, improving accuracy to **87%** and **F1-score** by **12%**, enabling data-driven poverty mitigation strategies.

Fritter - Twitter Clone **Jul–Dec'23**

- Led a team of three to develop a full-stack Twitter clone using **HTML**, **CSS**, **JavaScript**, **Python**, **PHP**, and **MySQL**, focusing on scalable architecture and secure user authentication.
- Implemented core Twitter features—tweets, comments, retweets, follows, and added voice messaging and time-limited post editing to enhance user engagement and reflect real-world platform behavior.